

Discussion 8

Def 1: Uniform Discrete Random variable

We say $X \sim \text{DisUnif}(\{a, a+1, \dots, b\})$ when Random variable X can only takes discrete values $a, a+1, \dots, b$ and they are all equally likely.

$$p_X(x) = \frac{1}{b-a+1}; \text{ where } x = a, a+1, \dots, b$$
$$E[X] = \frac{a+b}{2}$$
$$\text{Var}(X) = \frac{(b-a+1)^2 - 1}{12}$$

Eg1. A fair die is tossed twice, independently. Let the random variable Z be defined as the number of dots seen on the first toss minus the number of dots seen on the second toss. Find the expected value of Z^2 .

$$Z = X - Y, \quad \text{where}$$

$$X = Y = \text{Dis uniform}(\{1, 2, 3, \dots, 6\})$$

$$E[Z^2] = E[(X-Y)^2] = E[X^2 - 2XY + Y^2] = E[X^2] - 2E[XY] + E[Y^2]$$

$$\text{since } X \text{ and } Y \text{ are independent } E[XY] = E[X]E[Y]$$

$$E[Z^2] = E[X^2] - 2E[X]E[Y] + E[Y^2]$$

$$\text{since } X \text{ and } Y \text{ have same distribution } E[X] = E[Y] \text{ and } E[X^2] = E[Y^2]$$

$$E[Z^2] = 2E[X^2] - 2(E[X])^2 = 2(E[X^2] - (E[X])^2)$$

$$= 2 \text{Var}(X) \stackrel{\uparrow}{=} 2 \frac{(6-1+1)^2 - 1}{12} = \frac{35}{6}.$$

$X \sim \text{Disuniform}$
using formula

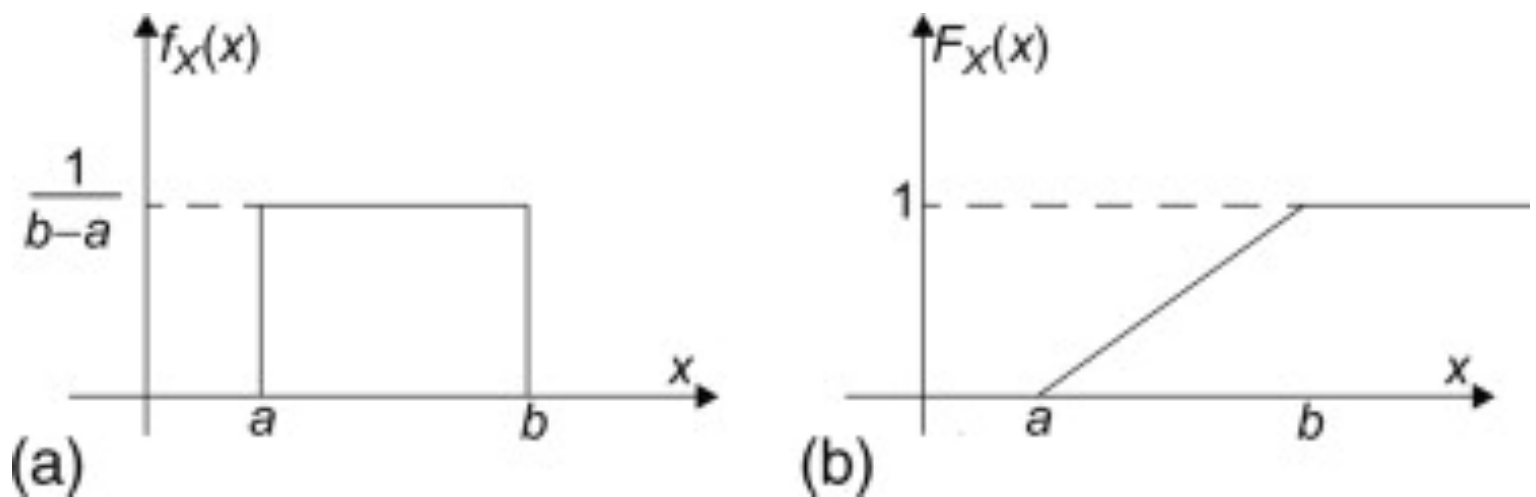


Figure 1: probability density function and cumulative distribution function of a uniformly distributed random variable

Def 2: Continuous Uniform Distribution $T \sim \text{Uniform}(a, b)$

The random variable X is uniformly distributed over the interval (a, b) when its probability density function is given by

$$f_T(t) = \begin{cases} \frac{1}{b-a} & \text{if } a < t < b \\ 0 & \text{otherwise} \end{cases}$$

It can be shown that

$$F_T(t) = \begin{cases} 0 & \text{if } t < a \\ \frac{t-a}{b-a} & \text{if } a \leq t < b \\ 1 & t \geq b \end{cases}$$

$$E[T] = \frac{a+b}{2}$$

$$\text{Var}(T) = \frac{(b-a)^2}{12}$$

Eg 2. Buses arrive at a specified stop at 15-minute intervals starting at 7 A.M. That is, they arrive at 7, 7:15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits

(a) less than 5 minutes for bus

(b) more than 10 minutes for a bus

Let T be the time that is uniformly distributed between $(0, 30)$, hence

$$f_T(t) = \begin{cases} \frac{1}{30} & 0 \leq t \leq 30 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned}
 (a) \quad P(10 < T < 15) + P(25 < T < 30) &= \int_{10}^{15} f_T(t) dt + \int_{25}^{30} f_T(t) dt \\
 &= \frac{15-10}{30} + \frac{30-25}{30} = \frac{10}{30} = \frac{1}{3}
 \end{aligned}$$

$$(b) \quad P(0 < T < 5) + P(15 < T < 20) = \frac{5}{30} + \frac{5}{30} = \frac{10}{30} = \frac{1}{3}$$

3. Let $X_1, X_2, \dots, X_{80}$ be a random sample of size $n = 80$ from a distribution with probability density function

$$f(x) = \begin{cases} 6x(1-x) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(0.48 < \bar{X} < 0.52)$. Hint: Begin by calculating μ and σ^2 .

$$\mu = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 6x^2(1-x) dx = \int_0^1 (6x^2 - 6x^3) dx = 2x^3 - \frac{3}{2}x^4 \Big|_0^1 = 2 - \frac{3}{2} = \frac{1}{2}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^1 6x^3(1-x) dx = \int_0^1 (6x^3 - 6x^4) dx = \frac{3}{2}x^4 - \frac{6}{5}x^5 \Big|_0^1 = \frac{3}{10}$$

$$\sigma^2 = E[X^2] - (E[X])^2 = \frac{3}{10} - \frac{1}{4} = \frac{6-5}{20} = \frac{1}{20}$$

$$E[\bar{X}] = E[X] = \frac{1}{2} = 0.5$$

$$Var(\bar{X}) = \frac{Var(X)}{n} = \frac{1}{1600} \text{ and } \sigma = \sqrt{Var(\bar{X})} = \frac{1}{40}$$

$$P(0.48 < \bar{X} < 0.52) = P\left(\frac{0.48 - 0.5}{\frac{1}{40}} < \frac{\bar{X} - E[\bar{X}]}{\sqrt{Var(\bar{X})}} < \frac{0.52 - 0.5}{\frac{1}{40}}\right) = P(-0.8 < \frac{\bar{X} - E[\bar{X}]}{\sqrt{Var(\bar{X})}} < 0.8) = \Phi(0.8) - \Phi(-0.8) \quad (1)$$

4. Use the Central limit theorem to approximate the following probabilities.

- (a) Suppose we roll a fair 7-sided die until we get 50 seven's. What is the probability it takes at least 500 rolls until this happens? (use continuity correction for this part.)
- (b) Let X be the sum of 500 real numbers, and Y be the same sum, but with each number rounded to the nearest integer before summing. If the fractions rounded off are independent and each one is uniformly distributed over $(-0.1, 0.1)$, use the Central Limit Theorem to estimate the probability that $P(|X - Y| > 30)$.

We know that $X \sim NBin(r = 50, p = \frac{1}{7})$ so $E[X] = \frac{r}{p} = 350$ and $var(X) = \frac{r(1-p)}{p^2} = 2100$. By the CLT $X \approx N(\mu = 350, \sigma^2 = 2100)$. We have to apply the continuity correction since we are approximating a discrete distribution (NBin) with a continuous one (Normal).

$$P(X \geq 500) = P(X \geq 499.5) = 1 - P(X < 499.5) = 1 - P\left(\frac{X - 350}{\sqrt{2100}} < \frac{499.5 - 350}{\sqrt{2100}} = \frac{149.5}{\sqrt{2100}}\right) = 1 - \Phi\left(\frac{149.5}{\sqrt{2100}}\right).$$

b. Let $\epsilon_1, \dots, \epsilon_n \in (-0.1, +0.1)$ be the roundoff errors ($n = 500$). Then,

$$E[\epsilon_i] = \frac{-0.1 + 0.1}{2} = 0$$

$$var[\epsilon_i] = \frac{(0.1 - (-0.1))^2}{12} = \frac{0.01}{3}$$

Let $X - Y = \epsilon = \sum_{i=1}^{500} \epsilon_i$, $E[\epsilon] = 0$ and $var(\epsilon) = \frac{5}{3}$. By the CLT, $\epsilon \approx N(\mu = 0, \sigma^2 = \frac{5}{3})$. Note we don't use a continuity correction here since we are approximating a continuous variable.

$$P(|\epsilon| > 30) = P(\epsilon > 30) + P(\epsilon < -30)$$

$$= P\left(\frac{\epsilon - 0}{\sqrt{\frac{5}{3}}} > \frac{30 - 0}{\sqrt{\frac{5}{3}}}\right) + P\left(\frac{\epsilon - 0}{\sqrt{\frac{5}{3}}} < -\frac{30 - 0}{\sqrt{\frac{5}{3}}}\right)$$

Let $Z = \frac{\epsilon - 0}{\sqrt{\frac{5}{3}}}$, then $Z \approx N(0, 1)$ and standard normal is symmetric about $Z=0$ hence $P(Z > \frac{30}{\sqrt{\frac{5}{3}}}) = P(Z < -\frac{30}{\sqrt{\frac{5}{3}}})$

$$P(|\epsilon| > 30) \approx 2P(Z > \frac{30\sqrt{3}}{\sqrt{5}}) = 2\left(1 - P(Z \leq \frac{30\sqrt{3}}{\sqrt{5}})\right) = 2(1 - \Phi(6\sqrt{15}))$$